



A
Project Synopsis
on
AI-Powered Legal Aid & Justice Accessibility System
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Notations / Nomenclature

Term / Acronym	Meaning
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
SDG-16	UN Sustainable Development Goal 16 (Peace, Justice & Strong Institutions)
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
MRR	Mean Reciprocal Rank
Hit@K	Proportion of queries for which a relevant source is found within the top K retrieved results
UI	User Interface
API	Application Programming Interface

Abstract

The AI-Powered Legal Aid & Justice Accessibility System aims to improve citizen access to accurate, source-grounded legal information under UN Sustainable Development Goal 16 (Peace, Justice and Strong Institutions). Rather than relying on a language model's memorized knowledge alone, the system retrieves relevant provisions from a curated Indian legal knowledge base before generating an answer, directly addressing the risk of hallucinated or unsupported legal responses.

The core of the project is a locally hosted, multilingual legal question-answering platform built to compare three configurations — an LLM-only baseline, dense Retrieval-Augmented Generation (RAG), and RAG with reranking — across English, Hindi, and Hinglish factual legal questions drawn from the Constitution of India and central Acts such as the Bharatiya Nyaya Sanhita, the Information Technology Act, and the Consumer Protection Act.

A reviewed, corpus-derived benchmark of 70 question groups (210 rows across the three languages) is used to measure factual accuracy, source-chunk retrieval (Hit@K, MRR), and generation reliability under identical, controlled conditions. Evaluation on this benchmark shows that retrieval substantially improves factual accuracy over the LLM-only baseline, while reranking further improves source recovery at a measurable latency cost. The system is designed as a pre-legal support tool — helping users identify relevant provisions and next steps — and is not intended as a replacement for a qualified legal professional.

Introduction

Access to justice remains a significant challenge in India due to high legal costs, language barriers, low legal literacy, and the technical, cross-referenced nature of statutory text. Citizens often cannot easily verify which Act or section actually governs their situation. Conversational AI built on Large Language Models (LLMs) can make legal text more approachable, but fluent answers are not automatically correct — an LLM may confidently state the wrong provision if it is not required to check a real source before answering.

Objective

- A large portion of society struggles to interpret legal terminology and locate the correct statutory provision.
- Language models can explain legal concepts fluently but may hallucinate incorrect sections, Acts, or citations if used without a verification mechanism.
- India's linguistic diversity means a legal-help tool must work across English, Hindi, and Hinglish, not English alone.
- In a legal context, a wrong or unsupported answer carries real consequences, so answers should be traceable back to an actual source.

- Retrieval-Augmented Generation offers a practical way to ground legal answers in real statutory text rather than a model's memorized knowledge.

Problem Gaps Identified

- Existing legal chatbots and government portals rarely verify whether their answers are grounded in an actual statute or section.
- Legal-AI evaluation is overwhelmingly English-only, leaving Hindi and Hinglish users largely unassessed.
- There is limited published, controlled comparison of plain LLM answering versus retrieval-augmented answering specifically for Indian statutory law.
- Government legal information portals are not conversational and assume the user already knows which Act or section to look up.
- Few systems separate “the retrieval found the right law” from “the model used it correctly,” which makes it hard to diagnose why a system gave a wrong answer.

Proposed Solution / Contribution

The proposed system addresses the above gaps through the following contributions:

1. Three-Condition Legal QA Platform — An LLM-only baseline, a dense-retrieval RAG configuration, and a reranked RAG configuration are implemented side by side under identical model and prompt settings, so that the effect of retrieval can be isolated and measured.
2. Multilingual Factual Question Benchmark — 70 reviewed question groups (210 rows) spanning English, Hindi, and Hinglish, covering constitutional, criminal, cyber, consumer, contract, property, and family-law questions with canonical answers and source labels.
3. Traceable Legal Knowledge Base — Statutory PDFs are ingested, chunked, embedded with a multilingual encoder, and indexed in a vector database so that every generated answer can be linked back to a specific provision.
4. Structured, Deterministic Evaluation — Factual accuracy, source-chunk Hit@K, Mean Reciprocal Rank, latency, and a stage-by-stage retrieval-failure analysis together diagnose where and why a system succeeds or fails.
5. Deployable Architecture — The retrieval-and-generation pipeline is exposed as a service that an existing legal-assistance web application can call, so the research prototype has a direct path to real usage rather than remaining a standalone script.

Technologies Used

- Frontend: React.js
- Backend / ML service: Python, FastAPI

- Local LLM runtime: Ollama, running the Qwen3 model family
- Embeddings: BGE-M3 (multilingual dense retrieval)
- Reranking: BGE-reranker-v2-m3
- Vector database: Qdrant
- Application database: MongoDB
- Dataset-generation aid: Gemini (used only to draft/translate candidate questions, all human-reviewed before use)
- Tools: Git/GitHub, Postman, Visual Studio Code

Field of the Project

Applied Natural Language Processing, Information Retrieval, and LegalTech — specifically, evaluation methodology for retrieval-grounded generative AI in a multilingual, factual, high-stakes domain.

Special Technical Terms

- Retrieval-Augmented Generation (RAG): Supplying a language model with retrieved external passages before it answers, instead of relying on memorized knowledge alone.
- Dense Retrieval / Semantic Search: Finding relevant passages by meaning, using vector embeddings, rather than exact keyword matching.
- Reranking: A second, more precise scoring pass applied to an initial set of retrieved candidates to improve their ordering.
- Groundedness: Whether an answer's claims are actually supported by the retrieved evidence.
- Hallucination: Model output that sounds plausible but is factually unsupported.
- Hit@K / MRR: Metrics for whether, and how highly, a relevant source appears among the top K retrieved results.
- Code-switching (Hinglish): Mixing Hindi and English within a single sentence or query, common in everyday Indian usage.

Literature Survey

A. Developments Between 2010 – 2015

During this period, legal information systems relied mainly on rule-based expert systems and keyword-based search. Retrieval approaches such as the BM25 ranking function [7] became the de facto standard for matching queries to documents by term overlap rather than meaning.

Key Developments:

- Rule-based legal expert systems that automated narrow tasks such as eligibility checks.
- Ontology-based legal knowledge bases for storing relationships between legal concepts.

Limitations:

- Systems could not interpret natural-language queries — only exact keywords or fixed rules.
- No semantic understanding: a query using different wording than the source document would simply fail to match.
- No mechanism for context-aware decision support.

B. Developments Between 2015 – 2020

This period saw a shift toward machine learning and the first neural retrieval models. Dense Passage Retrieval [8] showed that a learned dual-encoder could outperform traditional keyword search, and BERT-based rerankers [9] demonstrated that a second neural scoring pass could further improve which passages were surfaced first.

Key Developments:

- Machine learning classifiers (SVM, Naive Bayes, Logistic Regression) applied to legal document categorization.
- Dense, embedding-based retrieval began to replace purely keyword-based search.
- Early neural rerankers improved passage ordering after an initial retrieval pass.

Limitations:

- Machine learning models required large annotated datasets, which were scarce in the Indian legal domain.
- Chatbots remained largely rule-based, handling only predefined FAQs.
- Legal document summarization was unreliable given long sentences and dense terminology.

C. Developments From 2020 – Present

Transformer-based LLMs and the Retrieval-Augmented Generation paradigm [2] transformed what was possible: models could now retrieve external evidence and generate fluent, evidence-linked answers. Comprehensive surveys have since mapped this fast-moving area [10], while parallel work has documented the persistent hallucination problem in generative models [11].

In the Indian legal domain specifically, InLegalBERT [12] showed that further pretraining on Indian legal text improves downstream legal NLP tasks; the Indian Legal Documents Corpus (ILDC) [13] and the IL-TUR benchmark [14] established large-scale resources for judgment prediction and legal text understanding; and a recent evaluation of frontier LLMs on Indian legal reasoning [1] found that current models still struggle with long-form reasoning and citation discipline — directly motivating retrieval-grounded approaches such as the one built in this project. Multilingual embedding models [6] and modern open-weight LLMs [15] have made it practical to build such systems entirely on locally hosted infrastructure, while the code-switching literature [16] highlights why Hindi-English mixed queries (Hinglish) need explicit consideration rather than being treated as a variant of English. Standard IR evaluation practice, including heterogeneous retrieval benchmarks [17] and statistical significance testing for NLP results [18], informs how this project measures and reports its own findings.

Key Developments:

- Retrieval-Augmented Generation for grounding LLM answers in external documents.
- Domain-adapted legal language models (e.g., InLegalBERT) and large Indian legal benchmarks (ILDC, IL-TUR).
- Multilingual embedding and reranking models enabling cross-lingual semantic search.
- Structured, multi-metric evaluation of legal AI output, moving beyond simple accuracy.

Current Limitations (also observed in this project's own evaluation):

- Retrieval substantially reduces but does not eliminate factual errors — even the best configuration in this project's benchmark answers only a little over half of questions correctly.
- Recovering the correct document (e.g., the right Act) is noticeably easier than recovering the exact labelled passage within it.
- Hindi and Hinglish queries consistently lag behind English in both retrieval and answer accuracy.

- Small, single-reviewer benchmarks (such as the 70-group set used here) limit how far results can be generalized.

Methodology / Planning of Work

The methodology follows seven phases:

1. Requirement Analysis — Identify factual legal question types (provision identification and short-text lookup) across seven domains: constitutional, criminal, cyber, consumer, contract, property, and family law.
2. Dataset Preparation — Ingest statutory PDFs (the Constitution of India and relevant central Acts), clean and chunk the text with page/provision locators, generate candidate multilingual question groups, and have a reviewer verify canonical answers, aliases, and translations before freezing the test set.
3. Knowledge Base & Retrieval Pipeline Development — Convert chunks into vector embeddings with a multilingual encoder, index them in a vector database, and implement a dense-retrieval stage plus an optional reranking stage.
4. Model Integration — Connect a single, locally hosted LLM to two answering paths — an LLM-only path with no retrieved evidence, and a retrieval-augmented path — under identical generation settings so retrieval is the only variable that changes.
5. Evaluation Framework Development — Implement deterministic factual-match scoring, Hit@K and MRR computation for retrieval quality, end-to-end latency logging, and a stage-by-stage diagnosis of where a labelled source is lost (candidate search, ranking, or context budget).
6. Integration & Testing — Run a small pilot across all three conditions and all three languages to catch pipeline issues, then execute the full, frozen benchmark and record results, attempts, and retrieval traces for every question.
7. Final Deployment — Expose the retrieval-and-generation service to the existing legal-assistance web application for interactive use, alongside the standalone benchmark tooling used for this evaluation.

Facilities Required

Software Requirements

- Windows/Linux OS
- Python 3.x
- Node.js and React.js
- FastAPI (Python) for the retrieval/generation service
- MongoDB
- Qdrant (vector database)
- Ollama (local LLM runtime)

- Visual Studio Code, Postman, Git/GitHub
- Python libraries: sentence-transformers (or equivalent embedding library), NumPy, pandas

Hardware Requirements

- Laptop/workstation with a minimum of 16GB RAM
- i5/Ryzen 5 processor or higher
- GPU optional — the current configuration runs embedding and reranking on CPU; a GPU would reduce latency
- Stable internet connection (required for the external dataset-generation utility calls)

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